MILITARY SPECIFICATION

MOUNT, TELESCOPE: XM152

1. SCOPE

1.1 This specification defines the minimum requirements for one type of telescope mount which provides for boresighting adjustment in elevation and deflection.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel; Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F8625905	Fixture Final Inspection, Telescope Mount: XM152
F10516516	Mount, Telescope: XM52

PACKAGING DATA SHEET

MIL-P-14232/P10516516	Packaging of Mount, Telescope: XM152

FSC 1240

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(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3 Requirements.

3.1 First article testing.- Requirements for the submission of first article samples by the contractor for testing shall be as specified in the contract.

3.2 Fabrication.- The mount shall be manufactured in accordance with Drawing F10516516 and drawings pertaining thereto.

3.3 General specification.- The contractor shall be responsible for compliance with the following requirements of MIL-F-13926:

a. Order of precedence.
b. Dimensions and tolerances.
c. Inorganic protective surface finishes.
d. Part identification and marking.

3.4 Temperature conditions.- Except as specified in 3.6 and 3.7, the mount shall meet the requirements of this specification at a temperature between 60° and 90" Fahrenheit (F).

3.5 Workmanship.- The workmanship requirements of MIL-F-13926 shall be met. The exterior protective surface finishes shall be free of defects which may affect their protective qualities or appearance.

3.6 Extreme storage temperatures.- The mount shall show no evidence of physical failure due to storage at temperatures ranging from -80° to +160°F. Subsequent to meeting this requirement, the mount shall be capable of meeting the requirements of 3.8.

3.7 Extreme operating temperature.- The mount shall meet the applicable requirements of 3.15 while exposed and thermally stabilized at -40°F and +150°F.

3.8 Vibration.- This requirement shall be met with each boresight movement locked by means of the boresight locks. The mount shall show no evidence of physical failure due to being vibrated at 30 cycles per second and an amplitude of 0.06 inch, 0.12 inch total excursion (see 3.9). Subsequent to meeting this requirement, the mount shall be capable of meeting the following requirements of this specification.

3.9 Rigidity (line of sight retention).- The mount shall be capable of maintaining a previously established line of sight through the telescope objective tube assembly within 0.05 mil subsequent to being vibrated in accordance with 3.8.

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3.10 Telescope insertion and removal.- The mount shall be capable of permitting insertion and removal of the telescope objective tube assembly without restriction.

3.11 Orientation.- The requirements of 3.12 to 3.14.2 inclusive shall be met with the mount positioned as follows:

a. The flat mounting surface vertical within 15 minutes of arc.

b. The axis formed by the .3718 dowel hole and the .3723 slot at an angle of 38°9′16″ to the horizontal within 15 minutes of arc.

c. The axis through the center of the mounting surfaces of the two mounting rings at a 37°58′0′ angle with the axis formed by the .3718 dowel hole and the .3723 slot within .50 mil.

d. The axis through the center of the mounting surfaces of the two mounting rings parallel to the flat mounting surface within 0.50 mil.

3.12 Excursion ranges.

3.12.1 Elevation.- Rotation of the elevation knob shall cause the axis through the center of the mounting rings to elevate and depress a minimum of 20 mils from the position established in 3.llc.

3.12.2 Deflection.- Rotation of the deflection knob shall cause the axis through the center of the mounting rings to deflect a minimum of 20 mils in each direction from the position established in 3.lld.

3.13 Accuracy.

3.13.1 Elevation readings.- The elevation scale shall indicate 5.0 mils elevation of the axis through the center of the mounting rings within 0.25 mil excluding backlash. This requirement shall be met through the excursion range specified in 3.12.1.

3.13.2 Deflection readings.- The deflection scale shall indicate 5.0 mils deflection of the axis through the centers of the mounting rings within 0.25 mil excluding backlash. This requirement shall be met through the excursion range specified in 3.12.2.

3.13.3 Backlash.- Backlash in each boresight movement shall not exceed 0.05 mil.

3.14 Boresight locks.

3.14.1 Locking.- This requirement shall met with a 60-pound-inch (lb.-in.) load applied to each foresight locking lever to lock each boresight movement. There shall be no change in the position of the axis through the centers of the mounting rings when a 20 lb.-in. torque is applied to each boresight knob.

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3.14.2 Alignment detention. - The position of the axis through the centers of the mounting rings shall not change in excess of 0.05 mil due to locking and unlocking each boresight movement.

3.15 Operability of knobs, locks, and slip scales.

3.15.1 Boresight knobs.- The running torque required to rotate each boresight knob shall be within the limits of 2.0 and 10.0 lb.-ins. when checked at a temperature between 60° and 90°F. When checked at the temperature conditions specified in 3.7, the running torque shall not exceed 15.0 lb.-ha.

3.15.2 Boresight locks and slip scales.- The boresight locking levers and the slip scales shall be manually operable at a temperature between 60° and 90°F and the temperature conditions specified in 3.7.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection reequirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. -

4.1.1 General Provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing.- When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 Preproduction test.- Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3, 4.4 and 4.6 as well as any additional tests that may be specified in the contract.

4.2.2 Initial Production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

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4.2.3 First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as requied by the contract clauses applicable to first article testing.

4.3 Inspection provisions.

4.3.1 Submission of Product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurace Provision (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be perfomed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables 1, 11 and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

Class		REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 0.65% Defective		
101.	Vibration	3.8	4.6.3
102.	Rigidity, line of sight retention	3.9	4.6.4
103.	Telescope insertion and removal	3.10	4.6.5
104.	Elevation excursion range	3.12.1	4.6.6.1
105.	Deflection excursion range	3.12.2	4.6.6.2
106.	Elevation accuracy	3.13.1	4.6.6.3
107.	Deflection accuracy	3.13.2	4.6.6.4
108.	Backlash	3.13.3	4.6.6.5
109.	Boresight locks, locking	3.14.1	4.6.6.6
110.	Alignment retention	3.14.2	4.6.6.7

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TABLE I - CLASSIFICATION OF DEFECTS continued

	REQUIREMENT	TEST PROCEDURE
111. Boresight knobs, torque	3.15.1	4.6.7
112. Boresight locks, operability	3.15.2	4.6.8
113. Boresight slip scales, operability	3.15.2	4.6.8
114. Worknanship	3.5	4.6.9

MINOR: NONE

NOTE: The tests in Table I shall be conducted at a temperature between 60° and 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- Three mounts shall be selected at random by a Government representative as a special sample from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
301. Fabrication	3.2	Applicable drawings - visual
302. General specification	3.3	MIL-F-13926 - visual

4.4.2 Environmental.- Three mounts shall be selected at random by a Government representative as a special sample from each SO prduced or - from each month's production, whichever occurs first. The samples shall meet the requirements and tests in Table III and shall then meet the requirements and tests in Table I.

TABLE III

CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
303. Extreme storage temperature (-80° and +160°F)	3.6	4.6.1
304. Operability temperatures (-40° and +150°F)	3.7	4.6.2

6.

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4.4.3 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods- and procedures specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus,concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Performance and accuracy.- Special testing equipment conforming to Drawing F8625905 shall be utilized to test the following characteristics of the mount:

a. Rigidity, line of sight retention.
b. Telescope insertion and removal.
c. Elevation and deflection excursion ranges.
d. Elevation and deflection accuracy.
e. Backlash.
f. Boresight locks locking.
g. Alignment retention.

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4.6 Test methods and procedures.

4.6.1 Storage temperatures. - The test equipment used for this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.6. The mount shall be subjected to each temperature specified in 3.6 for a minimum of 6 hours. After the exposure to each temperature extreme, the mount shall be stabilized in an ambient temperature-between plus 60 degrees and 90 degrees Fahrenheit and given a visual and tactile examination for evidence of physical failure. The mount shall conform to the requirements of 3.6.

4.6.2 Extreme operating temperature.- The test equipment used for this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.7. The mount shall be subjected to each temperature extreme specified in 3.7 for a minimum of 6 hours. At the end of each 6 hour period and while still at the temperature extreme, the mount shall be subjected to the applicable tests specified in 4.6.7 and 4.6.8. The mount shall conform to the requirements of 3.7.

4.6.3 Vibration.- Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.8 shall be utilized for the vibration test. Unless otherwise specified the vibration test equipment shall be in conformance with the "Test Facilities" requirements of MIL-F-13926. Secure the Final Inspection Fixture F8625905 to the vibration table. Position the mount telescope in the fixture in accordance with the set-up instructions on Drawing F8625905. Establish the zero line of sight of the mount in accordance with Paragraph 1 of the fixture operating instructions on Drawing F8625905. The mount shall then be vibrated in a vertical plane for a period of 5 minutes $\pm$ 151 seconds. Immediately following the vibration cycle the mount shall be subjected to the test in 4.6.4. Subsequent to the test in 4.6.4, remove the adapter from the mount and subject the mount to a visual and tactile examination for evidence of physical failure such as: bent or broken parts, irregularities of moving parts, friction or looseness. The mount shall conform to the requirements of 3.8.

4.6.4 Rigidity (line of sight retention).- Perform this test in conjunction with the vibration test, in accordance with Paragraph 1 of . the fixture operating instructions on Drawing F8625905. The mount shall conform to the requirements of 3.9.

4.6.5 Telescope insertion and removal. conduct this test in sequence to the test in 4.6.4. Insert and remove the fixture adapter in the mount a minimum of three times. The mount shall conform to the requirements of 3.10.

4.6.6 Orientation.- The following tests shall be performed with - the mount- telescop~e positioned on the Inspection Fixture F8625905 in accordance with the set-up instructions on Drawing F8625905.

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4.6.6.1 Elevation excursion range.- Set the fixture Indicator “A” to zero. Rotate the elevation knob of the mount to its clockwise stop, then to its counterclockwise stop while observing the fixture Indicator "A". The elevation knob of the mount shall conform to the requirements of 3.12.1, as measured on Indicator “A”. Return the elevation knob to zero mils elevation at the completion of this test.

4.6.6.2 Deflection excursion range.- Set the fixture Indicator “C” to zero. Rotate the deflection knob of the mount to its clockwise stop, then to its counterclockwise stop while observing the fixture Indicator "C". The deflection knob of the mount shall conform to the requirements of 3.12.2, as measured on Indicator “C”. Return the deflection knob to zero mils deflection at the completion of this test.

4.6.6.3 Elevation accuracy.- Perform this test as in 4.6.6.1, except that backlash in the elevation knob movement shall be excluded for direction of travel. Rotate the elevation knob in 5 mil increments as read on the elevation mil scale of the knob. Each 5 mil rotation of the elevation knob as indicated on the elevation scale shall conform to the requirements of 3.13.1, as measured on the fixture Indicator “A” in accordance with Paragraph 3 of the fixture operating instructions on Drawing F8625905.

4.6.6.4 Deflection accuracy.- perform this test as in 4.6.6.3 utilizing the deflection knob and scale of the mount in conjunction with Indicator “C” of the fixture. Deflection knob accuracy of the mount shall conform to the requirements of 3.13.2, as measured on the fixture Indicator “C” in accordance with Paragraph 3 of the fixture operating instructions on Drawing F8625905.

4.6.6.5 Backlash.- The backlash test shall be performed concurrent with the tests in 4.6.6.3 and 4.6.6.4. As each new 5 mil increment reading on the elevation and deflection is made, read the appropriate indicator then advance the knob at least a half revolution and return it to the graduation on the knob (without overtravel), and again read the dial indicator. The difference in the two readings shall be considered backlash. The backlash in elevation and deflection shall not exceed the tolerance specified in 3.13.3, as measured on the appropriate fixture indicator in accordance with Paragraph 4 of the fixture operating instructions on Drawing F8625905.

4.6.6.6 Boresight locks, locking.- Perform this test in accordance with Paragraph 5 of the fixture operating instructions on Drawing F8625905. The boresight locks of the mount shall conform to the requirements of 3.14.1 as measured on Indicators “A” and “C” of the fixture.

4.6.6.7 Alignment retention.- Perform this test in accordance with Paragraph 6 of the fixture operating instructions on Drawing F8625905. Alignment retention of the mount shall conform to the requirements of 3.14.2 as measured on Indicators “A” and “C” of the fixtures.

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4.6.7 Boresight knobs, torque. - Standard torque measuring equipment capable of measuring the torque values specified in 3.15.1 shall be utilized for this test. An appropriate adapter shall be utilized with the torque measuring equipment to accommodate the boresight knobs of the mount. Torque shall be measured while rotating the boresight in each direction from its midpoint of travel. The running torque required to rotate the boresight knobs shall not exceed the applicable value specified in 3.15.1 for the temperature indicated.

4.6.8 Boresight locks and slip scales.- The boresight locks and slip scales shall conform to the requirements of 3.15.2, at the temperatures indicated, when subjected to a tactile examination.

4.6.9 Workmanship.- The mount shall be subjected to a visual and tactile examination, The mount shall conform to the requirements of 3.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10516516. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The XM152 Telescope Mount is intended for use in 105MM Gun Tank M60A1 to support the XM125 Articulated Telescope.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Applicable packaging data sheet (see 5.1).
(c) Selections of applicable levels of preservation, packaging, and packing.
(d) Applicable stock number.

Custodian:

Army-MU

Preparing activity:

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Project No. 1240-A462Y

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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